Practical-7

Problem Definition:

Design and implement a 4-bit synchronous counter that follows the non-linear custom sequence:

2 → 5 → 7 → 11 → 3 → 2 → … (repeating)

Requirements:

1. Draw the state diagram and state transition table.

2. Use any one type of flip-flop (JK, T, or D) for the design.

3. Derive flip-flop input equations using the respective excitation table.

4. Implement the complete synchronous circuit and verify the repeating sequence using simulation software or hardware.

Tools/Technology To Be Used:

- Flip-flops (D/JK/T)
- Logic gates
- Simulation software: Logisim / Multisim / Proteus / TinkerCad / Digital
- Breadboard & ICs (optional for hardware testing)

Aim:

To design and implement a 4-bit synchronous counter that follows a non-linear custom sequence (2 → 5 → 7 → 11 → 3 → 2 → …) using flip-flops and logic gates. The objective includes deriving the required flip-flop input equations using excitation tables and verifying the correct sequence through simulation or hardware implementation.

T flip flop:

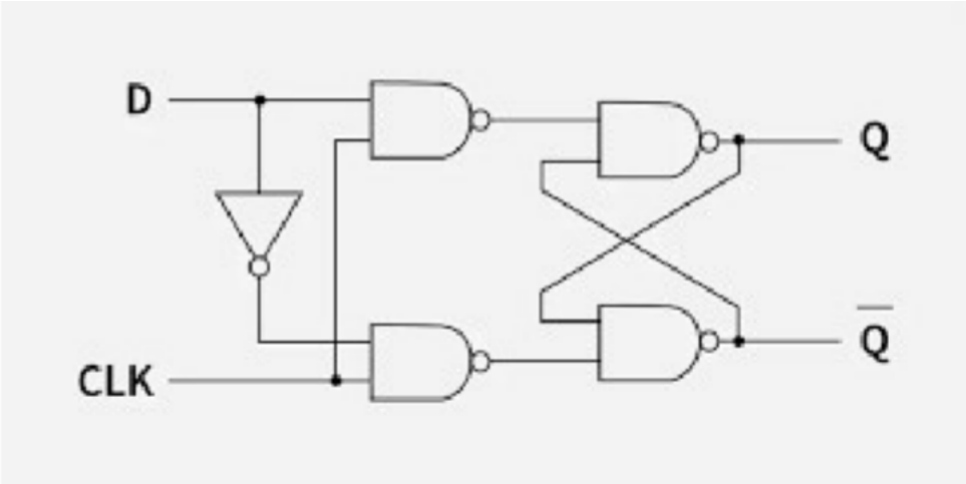
Truth table:

Q(n)	D(n)	Q(n+1) (Next)
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table:

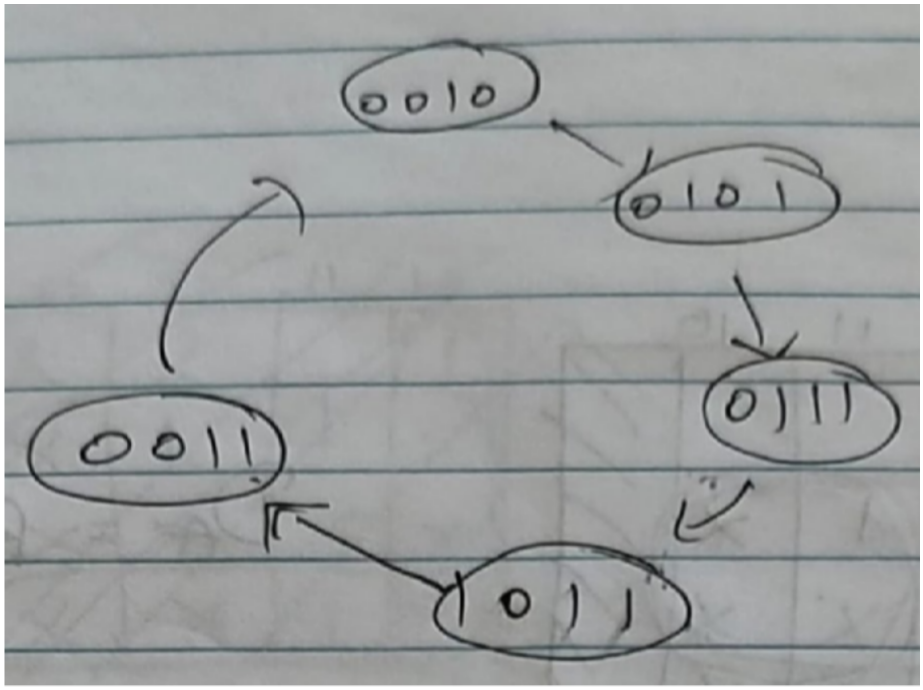
Q Output		Input
$Q_{(n)}$	$Q_{(n+1)}$	D_n
0	0	0
0	1	1
1	0	0
1	1	1

Circuit Diagram:



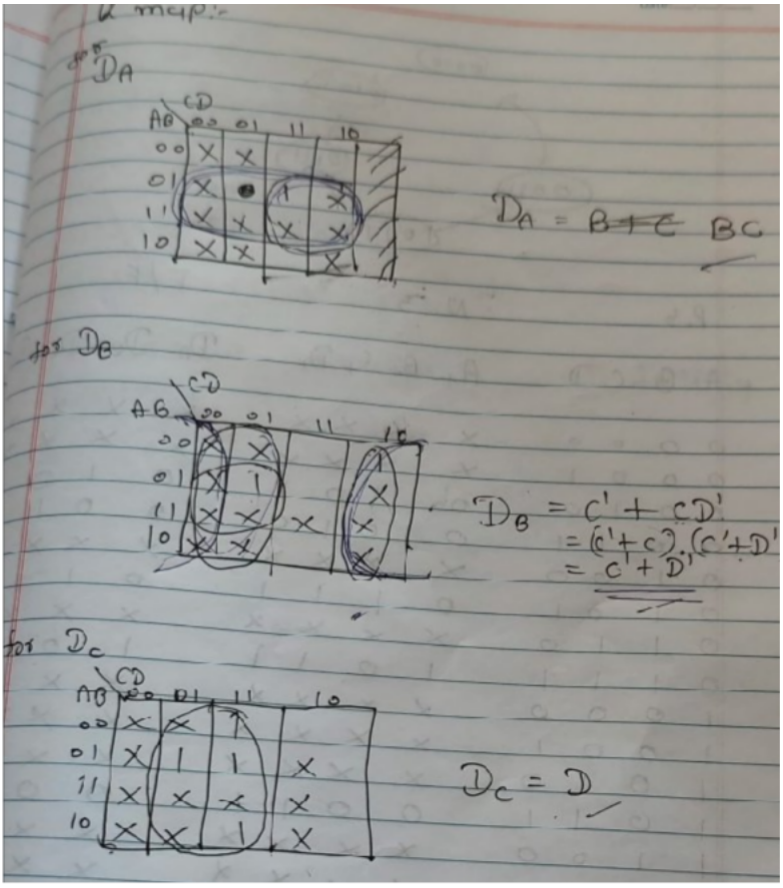
4-bit binary Random counter:

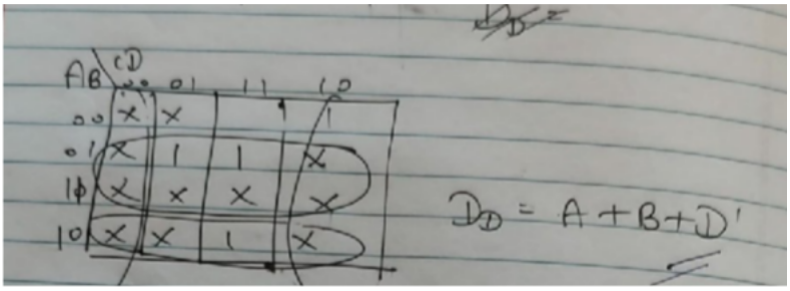
State diagram & State Table:



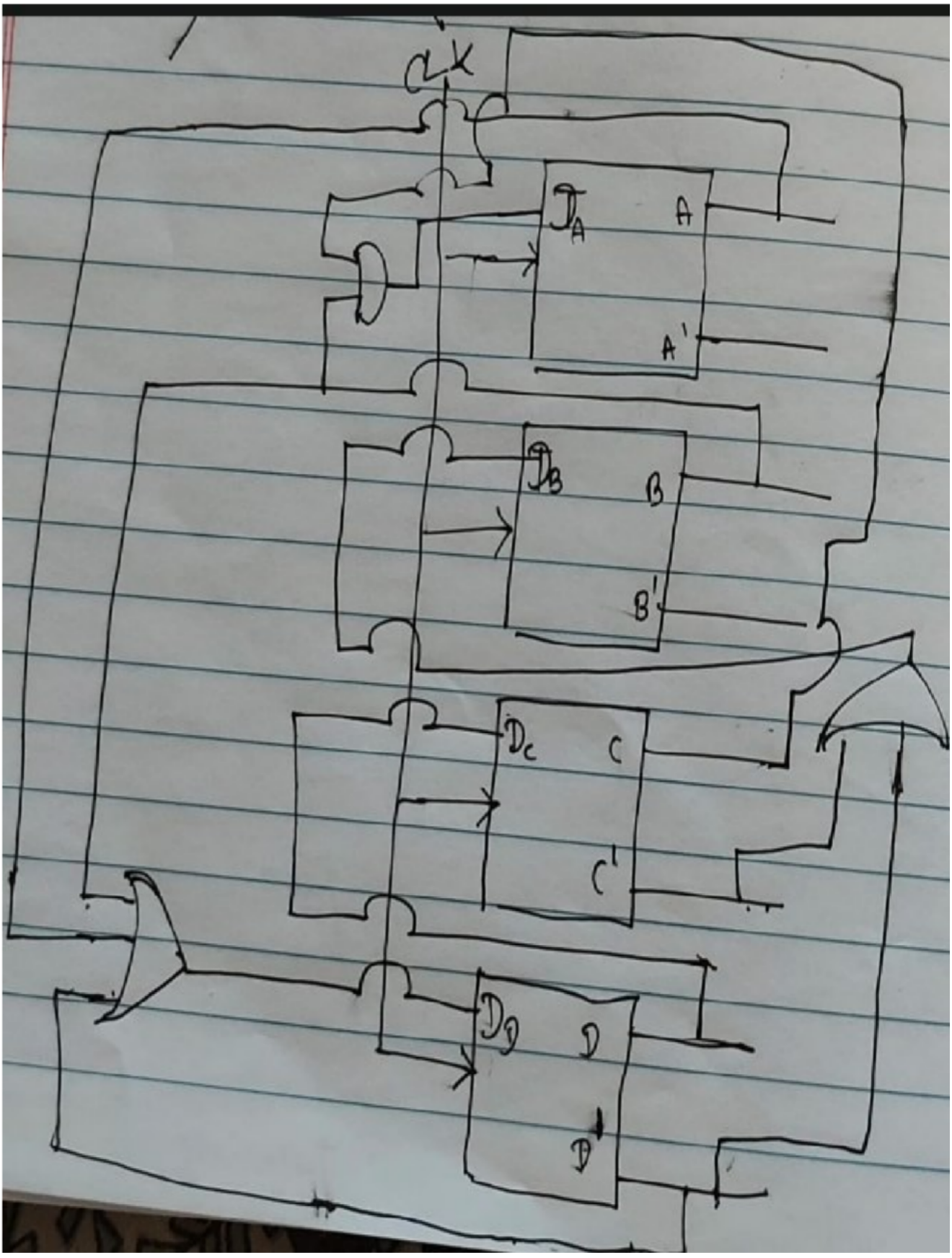
P.S				N. S				F/F			
A	B	C	D	A_+	B_+	C_+	D_+	D_A	D_B	D_C	D_D
0	0	0	0	x	x	x	x	x	x	x	x
0	0	0	1	x	x	x	x	x	x	x	x
0	0	1	0	0	1	0	1	0	1	0	1
0	0	1	1	0	0	1	0	0	0	1	0
0	1	0	0	x	x	x	x	x	x	x	x
0	1	0	1	0	1	1	1	0	1	1	1
0	1	1	0	x	x	x	x	x	x	x	x
0	1	1	1	1	0	1	1	1	0	1	1
1	0	0	0	x	x	x	x	x	x	x	x
1	0	0	1	x	x	x	x	x	x	x	x
1	0	1	0	x	x	x	x	x	x	x	x
1	0	1	1	0	0	1	1	0	0	1	1
1	1	0	0	x	x	x	x	x	x	x	x
1	1	0	1	x	x	x	x	x	x	x	x
1	1	1	0	x	x	x	x	x	x	x	x
1	1	1	1	x	x	x	x	x	x	x	x

K-Map & circuit diagram:

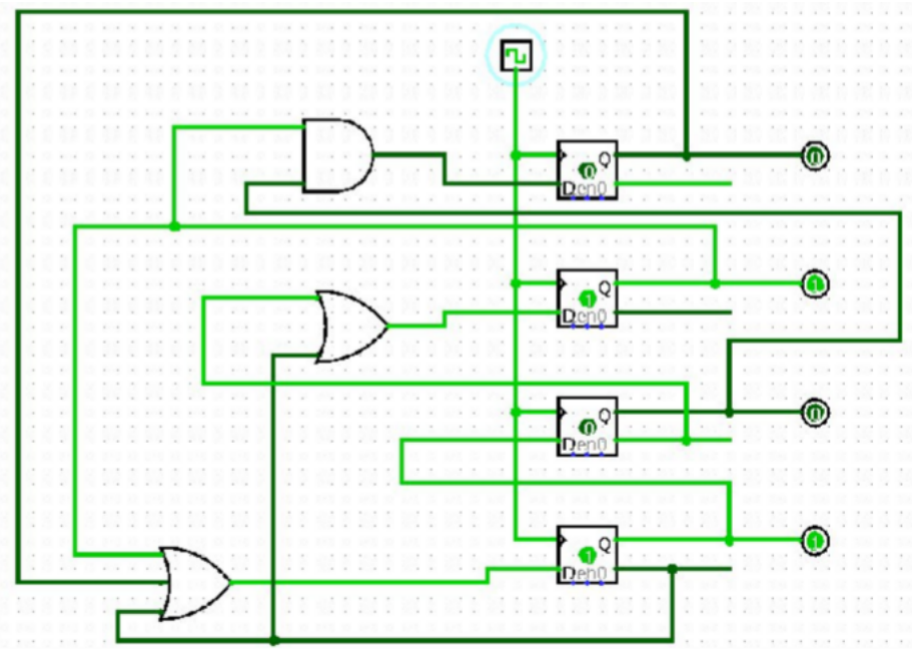


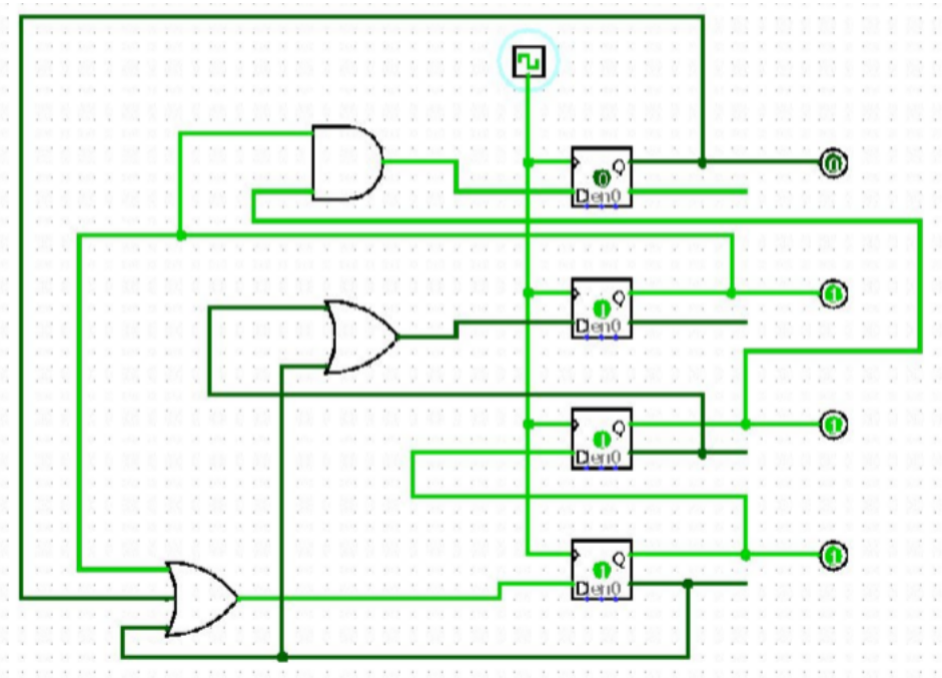


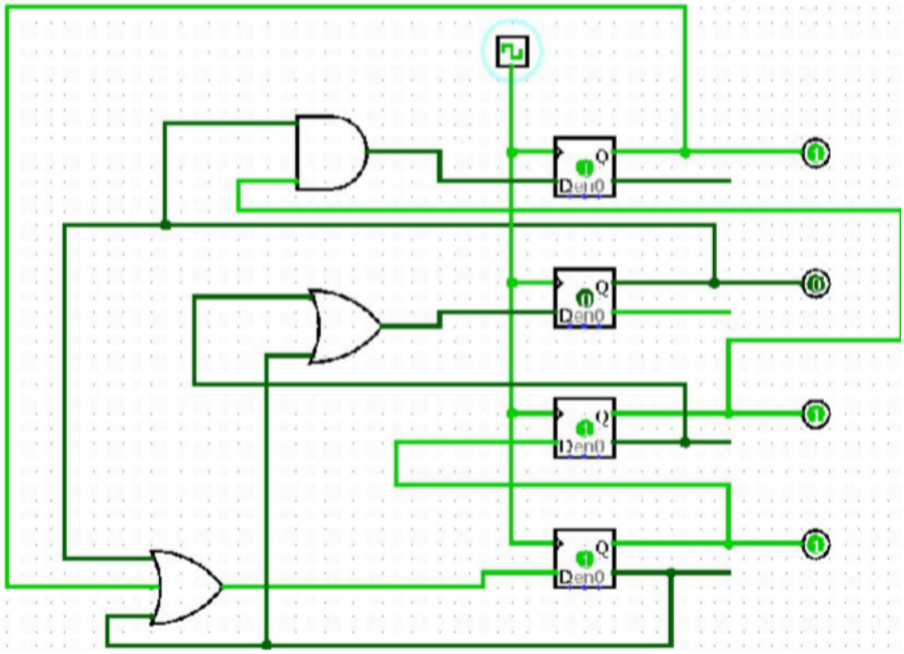
Circuit :

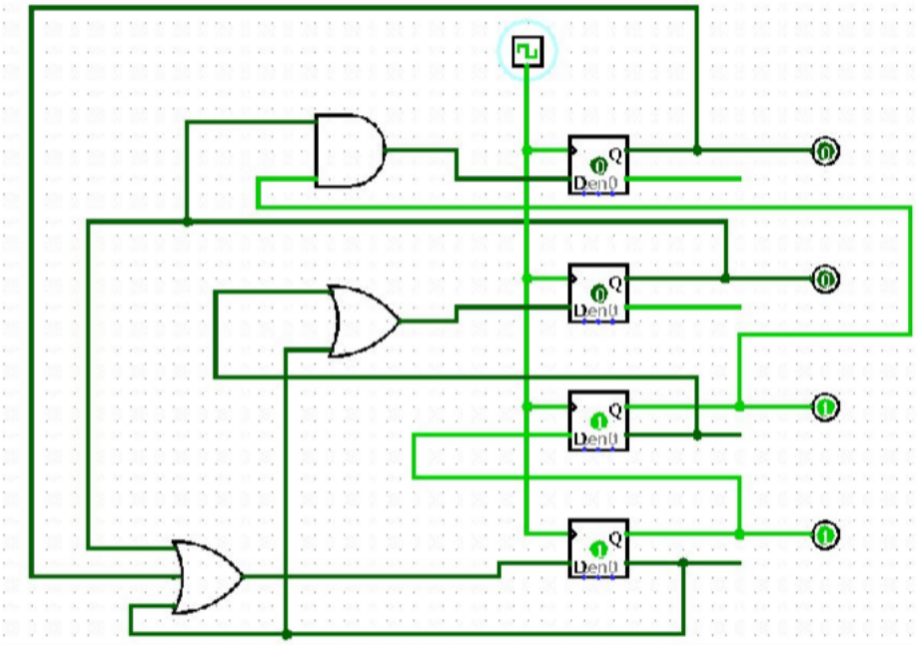


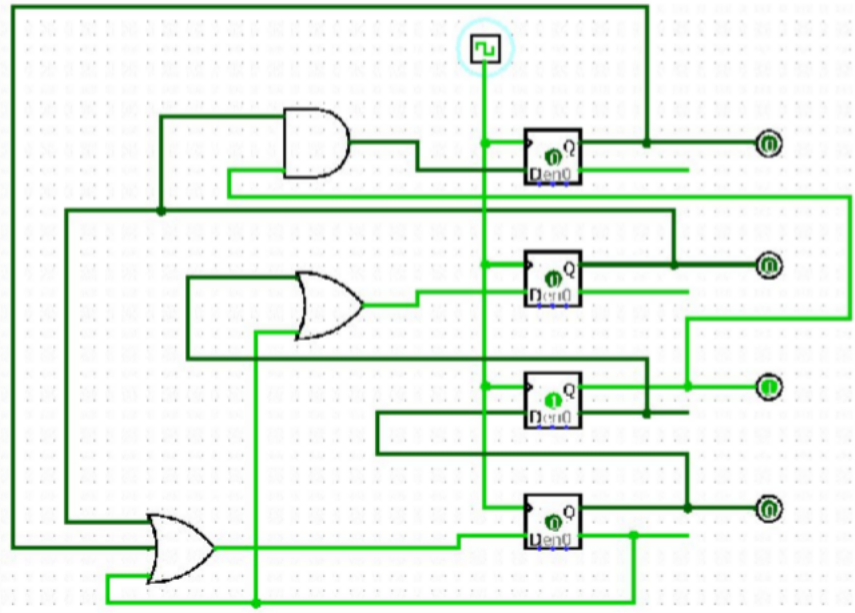
Screen shot:











Conclusion : This practical successfully demonstrates the design of a 4-bit synchronous counter tailored for a non-linear custom sequence. By utilizing state transition tables and flip-flop excitation logic, the circuit proves that synchronous systems can be programmed to skip traditional binary increments in favor of specific numerical patterns (2, 5, 7, 11, 3). The implementation confirms that by tying all flip-flop clock inputs to a common source, the system eliminates ripple delays and ensures all state changes occur simultaneously. Ultimately, the project validates the versatility of sequential logic design, showing how K-map minimization can transform a complex counting requirement into an optimized, hardware-efficient circuit.